Zhiqing (Angelika) Zhang

Mobile: +1-4753009976 | Email: angelikazhang966@gmail.com
Website: https://angelikazhang.github.io/ | LinkedIn: www.linkedin.com/in/az966

EDUCATION

Yale University

MPH Biostatistics 08/2025-05/2027

- **Notable Coursework:** Statistical Methods in Computational Biology; Applied Artificial Intelligence in Healthcare; Applied Machine Learning and Causal Inference; Longitudinal and Multilevel Data Analysis; Advanced Regression Models

Beijing Normal-Hong Kong Baptist University (BNBU)

BSc (Honours) Statistics (First Class) 09/2021-07/2025

- **GPA:** 3.72/4.0 | **Ranking:** 6/94
- **Notable Coursework:** Real Analysis; Survival Analysis; Data Mining; Time Series Analysis; Statistical Computing; Multivariate Analysis; Categorical Data Analysis; Experimental Design; Applied Stochastic Process; Network and Transportation Models; Advanced Statistics; Advanced Probability; Optimization; Linear Programming and Integer Programming; Regression Analysis; General Biology
- **Dissertation:** Prediction of response to chemotherapy via patient-derived tumor organoids test.

University of Oxford

Data Science Summer School (150 Hours) 08/2022

RESEARCH PROJECTS

Prediction of Response to Chemotherapy via Patient-derived Tumor Organoids 10/2023-12/2024

Supervisors: Ping He and Xiaoling Peng, BNBU

- Addressed heterogeneity in chemotherapy response using patient-derived tumor organoid data
- Designed and implemented experimental protocols and conducted statistical analysis on real biological samples
- Developed predictive models (regression + machine learning) to estimate treatment outcomes → enabled data-driven personalized therapy decisions
- Integrated biological insights with quantitative modeling to bridge experimental oncology and predictive analytics

Statistical Computing in Research 01/2024-08/2024

Supervisor: Divakaran Liginlal, Carnegie Mellon University

- Addressed high-dimensional challenges in cybersecurity datasets impacting model efficiency
- Applied statistical inference, regression modeling, and machine learning techniques
- Investigated synergy between dimensionality reduction and ML → improved detection robustness and efficiency
- Produced a comprehensive research report demonstrating methodological insights and performance improvements

Time Series Analysis 05/2024-06/2024

Supervisor: Bohai Zhang, BNBU

- Modeled weather data using SARIMA and LSTM approaches
- Evaluated performance via AIC, BIC, and Ljung-Box statistics → compared statistical vs deep learning models

Application of Logistic Regression in Airline Satisfaction 11/2023-12/2023

Supervisor: Xiaoling Peng, BNBU

- Assessed variable relationships using cross-tabulation and chi-square tests
- Built and interpreted logistic regression models to quantify service effects on customer satisfaction
- Identified on-time performance as the most significant predictor while highlighting service-related limitations

INTERNSHIP

Bayer China | Biostatistics Analyst 01/2024-05/2024

- Addressed pharmaceutical quality validation using hypothesis testing (z-tests, t-tests in R)
- Performed exploratory data analysis and statistical modeling on antihypertensive datasets
- Conducted survival analysis (Kaplan-Meier, log-rank) to evaluate treatment effects
- Developed deep learning models (CNN) for AI-assisted brain CT classification

International School, Guangzhou Nanfang College | Teaching Assistant 03/2025

- Delivered statistics tutoring and supported student comprehension
- Prepared instructional materials and contributed to academic seminars

Guangdong Kexinan Information Technology Co., Ltd | Network Security Intern 06/2022-07/2022

- Assisted in vulnerability assessment reporting and cybersecurity documentation
- Supported financial data analysis workflows and administrative operations

PUBLICATION

Zhang, Z., Chen, H. & Shen, J. Applying machine learning combined with dimensionality reduction to enhance intrusion detection. ICICR2025, May 16-18, 2025.

SKILLS

Technical: Python, C, R, Lingo, MATLAB, HTML, CSS

Platforms/Software: PLINK, MySQL, RStudio, PyCharm, CLion, Unity3D, OpenCV, Jupyter, Visual Studio Code, LaTeX, SPSS

Languages: Cantonese, Mandarin English, German

HONOURS & AWARDS

- Best Paper Award, ICICR2025 (2025)
- Successful Participant, Mathematical Contest in Modeling (2023, 2024)
- Merit Prize, China Undergraduate Mathematical Contest in Modeling (2023)
- Second Class Award Scholarship (2024)
- First Class Award Scholarship (2022, 2023)